

1.3 Virtualisation or Cloud Platform Protection

Control Type: Mandatory/Advisory for B	Applies to architecture:	A1	A2	A3	A4	B
		•	•	•	•	•
<u>Control Definition</u> Control Objective: Secure the virtualisation or cloud platform and virtual machines (VMs) that host Swift-related components to the same level as physical systems. In-scope components: • on-premises or remote (that is hosted or operated by a third party, such as a cloud provider, or both) virtualisation or cloud platform and VMs used to host any of the following Swift-related components: – messaging interface – communication interface – GUI – SwiftNet Link – Swift and customer (client) connector – customer client connector – jump server – dedicated and general-purpose operator PCs (when using virtual desktops) – firewalls – Alliance Connect Virtual, Alliance Connect Virtual on Public Cloud and Alliance Connect Virtual on Premises VPN instance – bridging servers (such as middleware or file transfer servers other than customer connectors) used for and guardian of the secure data exchange between back-office and Swift-related components Note: This requirement is not applicable when there is no on-premises and remote virtualisation or cloud platform and no VMs used to host the referred in-scope components. Note: It is reminded that containerised applications are by default considered as co-hosted on the same system that hosts the container itself. As new functionalities are incorporated into containers, when technically possible and supported by a risk assessment, separation and isolation, similar to virtualised machines, may be considered (see the CSP FAQ for some principles to incorporate as this is an alternative to the suggested guidelines). Risk Drivers: • unauthorised access • uncontrolled proliferation of systems and data						
<u>Implementation Guidance</u> Control Statement: Secure the virtualisation or cloud platform, virtualised machines, and the supporting virtual infrastructure (such as firewalls) to the same level as physical systems. Control Context: Security controls that apply to non-virtualised (physical) systems are equally applicable to virtual systems. The additional virtualisation layer needs extra attention from a security perspective. The uncontrolled proliferation of VMs could lead to unaccounted machines with the risk of unmanaged, unpatched systems open to unauthorised access to data. If appropriate controls have been implemented to this underlying layer, then Swift does not limit the use of virtual technology for any component, in the limit of its supportability, of the user's Swift infrastructure including virtual desktops, or the associated supporting infrastructure (for example, virtual firewalls). Implementation Guidelines:						

The implementation guidelines are common methods to apply the relevant control. The guidelines are a helpful way to begin an assessment but should never be considered as an audit checklist as each user's implementation may vary. Therefore, in cases where some implementation guidelines elements are not present or partially covered, mitigations as well as particular environment specificities must be considered to properly assess the overall compliance adherence level (as per the suggested guidelines or as per the alternatives).

- The same security requirements apply to the virtualisation or cloud platform, virtual machines, and supporting virtual infrastructure as for all other infrastructure systems and components. Those security requirements cover, for example, the location in an existing secure zone that has similar controls as those applicable to the Swift or customer secure zone, privileged access restrictions, login and password policies, installation of security updates, and restriction of internet access. Those controls have the virtualisation or cloud platform identified in the *In-scope Components* section.
- Vulnerability scanning is performed on Swift-related VMs and, when technically possible, on the virtualisation or cloud platform.
- The virtualisation or cloud platform systems are subject to physical protection, which prevents unauthorised physical access.
- VM isolation is ensured on the virtualisation or cloud platform to prevent the lateral move out of a virtual machine to access or interact with other VMs (or the underlying hypervisor) or to bypass normal network controls that filter or inspect connections to the Swift environment (or a combination of both).
 - Filtering and expected inspections of the network flows that reach the Swift-related VMs are performed preferably using resources (such as firewalls, packet inspections, or content filtering) external to the virtualisation platform or must be enforced at the hypervisor level.
 - If isolation is ensured on the virtualisation or cloud platform, then the hosted VMs can maintain their security classification and can be individually secured accordingly (as such, they do not inherit the classification of the Swift-related VMs and are not subject to all Swift-related controls).
- When multi-factor authentication is implemented for interactive access to the Swift-related VM operating systems (and in-line with control 4.2) to also prevent direct access to those VMs from the hypervisor layer, then multi-factor authentication is not mandatory at the virtualisation platform management level.
- When relying on a third party for the underlying virtualisation or cloud platform, the user must engage with the third party to obtain reasonable comfort that the control objective is met.

Optional Enhancement:

- Apply the control to all bridging servers (such as middleware or file transfer servers other than customer connectors) used for data exchange between back-office and Swift-related components.